Anaemia — Diagnosis and Management

1. Definition and Epidemiology

Anaemia is defined by the WHO as haemoglobin (Hb) below: 13 g/dL in men ≥15 years; 12 g/dL in non-pregnant women ≥15 years; 11 g/dL in pregnant women; 11 g/dL in children 6 months-5 years; 11.5 g/dL in children 5-11 years; 12 g/dL in children 12-14 years.

India has one of the highest burdens of anaemia globally. The National Family Health Survey 5 (NFHS-5, 2019-21) found 57% of children under 5, 57% of women aged 15-49, and 25% of men aged 15-49 were anaemic. Anaemia in pregnant women (50.4%) remains extremely high.

Iron deficiency anaemia (IDA) is the most common cause, followed by anaemia of chronic disease, vitamin B12 and folate deficiency, haemoglobinopathies (sickle cell disease, thalassaemia — very prevalent in tribal India), and haemolytic anaemias.

2. Classification and Approach

MCV-Based Classification	Causes
Microcytic (MCV <80 fL)	Iron deficiency, Thalassaemia, Sideroblastic anaemia, Anaemia of chronic disease (u
Normocytic (MCV 80-100 fL)	Anaemia of chronic disease, Aplastic anaemia, Haemolytic anaemia, Acute blood los
Macrocytic (MCV >100 fL)	B12 deficiency, Folate deficiency, Hypothyroidism, Alcohol, Liver disease, Drug-induc

The reticulocyte count distinguishes hypoproliferative (low reticulocytes — bone marrow not responding: IDA, B12/folate deficiency, aplastic anaemia, CKD) from hyperproliferative anaemia (elevated reticulocytes — bone marrow responding: haemolysis, acute blood loss).

3. Iron Deficiency Anaemia

Causes: inadequate dietary intake (vegetarians, infants on exclusive milk diet), malabsorption (coeliac disease, post-gastrectomy, Helicobacter pylori), increased demand (pregnancy, infancy, adolescence), blood loss (menorrhagia, GI bleeding from NSAIDs/PUD/cancer, hookworm infestation — very common in rural India).

Laboratory findings: low Hb, low MCV (microcytic), low MCH (hypochromic), low serum ferritin (<12 mcg/L confirms IDA), low serum iron, high TIBC, low transferrin saturation (<16%), elevated RDW (anisocytosis).

Treatment: identify and treat the cause. Ferrous sulphate 200 mg TDS (providing ~195 mg elemental iron/day) between meals. GI side effects common (constipation, nausea) — take with orange juice (enhances absorption), avoid tea/coffee (inhibit absorption). Check Hb after 4 weeks (rise of 1-2 g/dL expected). Continue for 3-6 months after Hb normalises to replenish stores.

IV iron (ferric carboxymaltose, iron sucrose): for intolerance to oral iron, malabsorption, non-compliance, rapid correction needed (pre-operative, pregnancy), or CKD patients. Highly effective single-infusion options now available.

4. Vitamin B12 and Folate Deficiency

B12 deficiency: megaloblastic anaemia + neurological manifestations (subacute combined degeneration of the cord — dorsal and lateral column dysfunction: peripheral neuropathy, loss of vibration sense, proprioception, upper motor neuron signs). Psychiatric manifestations possible.

Causes of B12 deficiency: pernicious anaemia (autoimmune — anti-intrinsic factor antibodies, most common in older adults), strict vegetarianism (B12 only in animal products — India has very high prevalence among vegetarians), malabsorption (terminal ileum disease — Crohn's, ileal resection), metformin (reduces B12 absorption — monitor in long-term users).

Treatment: hydroxocobalamin IM injections — 1 mg IM on alternate days for 2 weeks, then 1 mg IM every 3 months lifelong (if pernicious anaemia or irreversible malabsorption). High-dose oral cyanocobalamin 1000-2000 mcg daily is an alternative where compliance allows (passive absorption bypasses intrinsic factor).

Folate deficiency: similar blood picture to B12 but no neurological features (important differential). Causes: poor diet (green vegetables, pulses), malabsorption, alcohol, pregnancy (increased demand), antifolate drugs (methotrexate, trimethoprim). Treatment: folic acid 5 mg daily. IMPORTANT: never treat folate without ruling out B12 deficiency first — folate supplementation can mask B12 deficiency and allow neurological damage to progress.

5. Anaemia of Chronic Disease

Anaemia of chronic disease (ACD), also called anaemia of inflammation, is the most common anaemia in hospitalised patients. Associated with: chronic infections (TB, HIV), inflammatory conditions (RA, SLE, IBD), malignancy, CKD.

Pathophysiology: hepcidin (acute phase reactant) is upregulated by IL-6 during inflammation → blocks iron absorption and ferroportin-mediated iron release from stores → functional iron deficiency + reduced EPO response.

Features: normocytic or mildly microcytic, low serum iron, low TIBC (differentiates from IDA where TIBC is high), normal or high ferritin (ferritin is also an acute phase protein — can be normal even with true IDD). Soluble transferrin receptor (sTfR) and sTfR-F index help distinguish from IDA.

Treatment: primarily treat the underlying cause. IV iron in CKD + ESAs (erythropoietin) for ACD in CKD. Red cell transfusion for symptomatic severe anaemia (Hb <7-8 g/dL) or acute decompensation.

6. Haemolytic Anaemias

Haemolytic anaemia: shortened red cell survival (<120 days). Compensated: bone marrow increases output to match destruction. Uncompensated: anaemia results.

Intravascular haemolysis: within blood vessels. Causes: autoimmune haemolytic anaemia (AIHA, cold), G6PD deficiency triggered by oxidant stress (drugs, fava beans, infections), transfusion reactions, mechanical haemolysis (prosthetic heart valve). Features: haemoglobinaemia, haemoglobinuria (dark red urine), very low haptoglobin, elevated LDH.

Extravascular haemolysis: spleen removes damaged RBCs. Causes: autoimmune (warm AIHA — IgG, DAT positive), hereditary spherocytosis (HE), sickle cell, thalassaemia. Features: splenomegaly, jaundice (unconjugated), raised urinary urobilinogen, elevated LDH.

7. Sickle Cell Disease and Thalassaemia — India

Sickle cell disease (SCD): autosomal recessive haemoglobin disorder (HbSS or HbSC). Affects tribal communities of Chhattisgarh, Madhya Pradesh, Odisha, Maharashtra, and Gujarat. Estimated 200,000-300,000 SCD patients in India.

Acute sickle crises: vaso-occlusive pain crisis (severe bone/joint pain), acute chest syndrome (ACS — fever, respiratory symptoms, infiltrate on CXR — can be fatal), stroke, splenic sequestration, priapism. Management: IV hydration, analgesia, oxygen, transfusion for ACS/stroke/severe anaemia.

Long-term SCD management: hydroxyurea (increases HbF, reduces crises), penicillin V prophylaxis (prevents pneumococcal sepsis in hyposplenic patients), folate supplementation, pneumococcal and meningococcal vaccination, regular transcranial Doppler (TCD) screening for stroke risk in children.

Thalassaemia: beta-thalassaemia major — transfusion-dependent anaemia from childhood. Prevalent in Gujarat, Punjab, Maharashtra, and Sindhi communities. Treatment: regular red cell transfusion + iron chelation (desferrioxamine, deferasirox). Curative: bone marrow transplant from matched sibling donor. Gene therapy (betibeglogene — FDA approved 2022) is emerging.

National Health Mission includes thalassaemia screening and counselling. Premarital/prenatal carrier testing prevents new thalassaemia major births.

8. Aplastic Anaemia

Aplastic anaemia: pancytopenia (low RBC, WBC, and platelets) due to failure of the bone marrow to produce blood cells. Bone marrow biopsy shows hypocellular marrow.

Causes: most commonly idiopathic autoimmune (immune-mediated destruction of HSCs). Secondary: drugs (carbamazepine, chloramphenicol, NSAIDs), hepatitis (seronegative aplastic anaemia post-hepatitis), radiation, viral infections (EBV, CMV, parvovirus B19), PNH (paroxysmal nocturnal haemoglobinuria — related).

Severe aplastic anaemia (SAA): neutrophils <0.5×10■/L + platelets <20×10■/L + reticulocytes <60×10■/L. Treatment: matched sibling donor BMT (young patients) — curative. IST (immunosuppressive therapy): anti-thymocyte globulin (ATG) + ciclosporin + eltrombopag for patients without sibling donor or older patients. Transfusion support.

9. Anaemia in Pregnancy

Anaemia in pregnancy (Hb <11 g/dL) affects 50% of Indian pregnant women and is a major contributor to maternal and perinatal mortality. Physiological haemodilution in pregnancy: plasma volume increases more than red cell mass → relative fall in Hb (lowest at 28-32 weeks).

Prevention and treatment: Iron-Folic Acid (IFA) supplementation — all pregnant women (Iron 60 mg + Folic acid 400 mcg daily throughout pregnancy and 3 months post-partum) under India's WIFS programme. Treat severe anaemia (Hb <7 g/dL) with IV iron or blood transfusion before delivery.

Anaemia in pregnancy increases risk of: maternal death, cardiac failure, infections, pre-eclampsia, low birth weight, preterm birth, perinatal mortality. National Anaemia Mukt Bharat initiative targets anaemia across lifecycle.

10. Anaemia Prevention in India — Public Health Measures

Nutritional interventions: IFA supplementation for adolescent girls (WIFS — Weekly Iron and Folic Acid Supplementation Programme) and pregnant women. Mid-Day Meal Programme with iron-rich foods (green leafy vegetables, legumes) for schoolchildren.

Deworming: albendazole 400 mg every 6 months for children and adolescents reduces hookworm infestation — a major contributor to IDA in India. National Deworming Day (February 10, August 10 in select states) treats over 100 million children annually.

POSHAN Abhiyaan (National Nutrition Mission): targets reduction of stunting, wasting, anaemia, and low birth weight. Nutritional rehabilitation centres for severe acute malnutrition.

Food fortification: mandatory fortification of edible oil with Vitamin A and D, milk, wheat flour, and rice is being scaled up under FSSAI regulations. Double-fortified salt (DFS — iron + iodine) has been shown effective in reducing IDA.